

Portfolio-Conditioned Dynamic Weakness and Reinforcement Ranking in Inverter-Rich Power Networks

Jorge L. Mayorga Taborda

Abstract—Weak-bus and grid-strength indices rank locations once, for a given network. When synchronous generators (SGs) are replaced by grid-following converters (GFLs) unit by unit, the quantity a planner needs is the marginal small-signal effect of the next replacement, and this effect depends on which units have already been replaced. On the IEEE 39-bus system with a documented GFL model, we evaluate all 2^9 replacement portfolios at 63 controller policies, 24 of them a new preregistered holdout, and measure the change of the transverse spectral abscissa $\alpha_{\perp}$ caused by each replacement in each portfolio. In 17/19 base-stable new-holdout policies the same replacement moves the same tracked electromechanical mode left in one stable portfolio and right in a portfolio that contains it. By the chain identity for set-function marginals, each such nested reversal is certified by a discrete-curvature term of the required sign, so $\alpha_{\perp}$ is neither submodular nor supermodular. An oracle fixed ranking of units orders 0.75 of resolved pairs and is materially wrong in 0.38 of next-replacement decisions, a failure carried mostly by transitions into fast converter-control modes; restricted to electromechanical transitions it is correct for 0.97 of pairs. For branch reinforcement, the portfolio-conditioned re-equilibrated eigenvalue sensitivity ranks finite reinforcements with median Spearman 0.985 on preregistration-eligible held-out conditions (0.987 over all 64), against 0.491 for the discovery-selected static index and 0.499 for the best of nine static indices chosen per condition. A frozen WECC library GFL chain reproduces the ranking method but not the reversals or the ranking itself. Replacement planning should therefore rank interventions in the portfolio context in which they are applied.

Index Terms—Grid-following converters, small-signal stability, eigenvalue sensitivity, grid strength, submodularity, replacement planning, weak buses.

I. INTRODUCTION

GRID-STRENGTH indices summarize how well a network supports converter-interfaced resources at a location. The short-circuit ratio (SCR) and its multi-infeed generalizations—the site-dependent SCR [1], the generalized SCR [2]–[4], the generalized operational SCR [5] and small-signal strength metrics for all-inverter systems [6]—and impedance-based metrics [7], [8] are used to flag weak buses and to rank where converters or reinforcements should go. Modal participation analysis [9], [10] and impedance-based eigenvalue sensitivities [11] play the same role for dynamic criticality.

These tools answer a question about a fixed system. Retirement of synchronous generators poses a different question.

J. L. Mayorga Taborda is with the Department of Electrical and Electronic Engineering, Universidad de los Andes, Bogotá, Colombia (e-mail: jl.mayorga@uniandes.edu.co).

Units are replaced one at a time, and the planner must decide which unit to replace next *given* the units already replaced, and which branch to reinforce *given* the resulting portfolio. The object of interest is the contextual marginal

$$\Delta_i \alpha_{\perp}(S) = \alpha_{\perp}(S \cup \{i\}) - \alpha_{\perp}(S), \quad (1)$$

the change of the rightmost transverse eigenvalue's real part when unit i is replaced in portfolio S . If $\Delta_i \alpha_{\perp}(S)$ kept its sign and its order across portfolios, a fixed ranking of units (a weak-bus list) would be adequate for sequencing. We show on the IEEE 39-bus system that it does not, and we characterize where the failure comes from.

Several ingredients are known and are credited here rather than claimed. The effect of a branch reinforcement on linear stability can have either sign (Braess-type behavior in oscillator networks [12] and in flows [13]; operating-point-dependent edge weights in [14]). The effect of adding a unit type can depend on the existing mix [15]–[17]. Eigenvalue sensitivities that account for the re-solved operating point are classical [18], [19] and have been applied to redispatch [20], operational parameters [21] and, in a recent preprint, line admittances [22]. Transmission switching changes small-signal margins [23], [24]. Submodular structure is standard in actuator placement [25], and its failure for natural dynamic set functions is known [26].

Within a focused 2020–2026 search (Section VII) we found no study that (i) certifies sign reversals of the same SG→GFL replacement across *nested* stable portfolios on the *same tracked electromechanical mode*, (ii) measures how often an oracle-optimal fixed ranking selects a materially worse next replacement, and (iii) validates a portfolio-conditioned re-equilibrated branch sensitivity as a *ranking* of finite reinforcements against static indicators on held-out conditions. The contributions are:

- C1) *Nested contextual sign reversal*. The same replacement moves the same electromechanical mode in opposite directions in two nested stable portfolios, in 17/19 base-stable policies of a new preregistered holdout. The classical chain identity certifies each reversal by a discrete-curvature term, so $\alpha_{\perp}$ is neither submodular nor supermodular.
- C2) *Fixed-ranking insufficiency and its modal anatomy*. The best fixed ranking of units is materially wrong in 0.38 of stable next-replacement decisions and does not transfer across policies. The failure is carried by transitions into

TABLE I
DEVICE AND BENCHMARK DATA (PER UNIT ON DEVICE RATING)

SG	two-axis, $D = 0$, constant P_m ; AVR $K_A = 10.1$, $T_E = 0.25$ s; PSS $K_S = -2$, $T_5 = 1$, $T_6 = 4.2$, $T_4 = 0.75$ s
PLL	$k_{p,\text{pll}} = 53$, $k_{i,\text{pll}} = 1400$
GFL loops	$\tau = 0.03$ s; outer 0.2/8; inner 0.25/6; $x_f = 0.15$, $r_f = 0.01$
Q/V	$k_{p,v}/k_{i,v} = 2/20$, leak $w = 0.05$ rad/s, gain $g \in [0, 1]$
Candidates	$V_9 = \{30, \dots, 38\}$ (2 ⁹ portfolios); core $V_4 = \{30, 33, 35, 37\}$

fast converter-control modes; restricted to electromechanical transitions a fixed ranking orders 0.97 of pairs.

- C3) *Portfolio-conditioned reinforcement ranking.* The re-equilibrated first-order sensitivity of $\alpha_\perp$ to branch admittance ranks finite reinforcements far better than nine static indicators on held-out policies and parameter draws, and the re-equilibration term, zero for controller coordinates, is what separates it from the frozen derivative.
- C4) *Electromechanical topology effects.* Single admissible branch actions both remove and create electromechanical incompatibilities, and seven static topology scores fail a preregistered prediction rule.

A cross-model holdout with a frozen WECC library GFL chain delimits these claims: the ranking method transfers, the reversals and the ranking itself do not. All thresholds, holdout points and tests were frozen before the computations reported here; deviations are logged. The study is phasor-domain small-signal analysis on one benchmark; no electromagnetic-transient claim is made.

II. PROBLEM FORMULATION

A. Model

The network is the IEEE 39-bus system [27] (46 branches, ten SGs, bus 39 an interconnection equivalent) in phasor form, with constant-power loads. The system is the index-one DAE $\dot{x} = f(x, z; p)$, $0 = g(x, z; p)$, where x collects device states, z the rectangular bus voltages and p parameters. At an equilibrium (x^*, z^*) the reduced Jacobian is $A = f_x - f_z g_z^{-1} g_x$.

Each SG uses a two-axis model on its own base [28] with $D = 0$, constant mechanical power, a first-order AVR $T_E \dot{e}_{fd} = K_A(v_{\text{ref}} + v_s - |V|) - e_{fd}$ and a power-input stabilizer. Each GFL has a synchronous-reference-frame PLL, filtered power measurements, PI outer and inner loops behind a filter reactance, and a leaky Q/V regulator $q_{\text{cmd}} = k_{p,v} g e + x_v$, $\dot{x}_v = k_{i,v} g e - w(x_v - q_{\text{ref}})$, $e = v_{\text{ref}} - |V|$, $w = 0.05$ rad/s. Parameters are those of the harmonized benchmark in Table I. Converter current limits, DC links and modulation delays are not modeled.

A policy $\theta = (g, k, t, h)$ sets the Q/V gain g of every GFL, a common scale k of the surviving SGs' AVR gains, a scale t of their AVR time constants, and a heterogeneity exponent h with $T_{E,i} \mapsto G(T_{E,i}/G)^h$, G the geometric mean. Replacing unit i installs a GFL with the same active and reactive dispatch and the SG rating (matched dispatch), so the AC power flow is the same for every portfolio.

B. Transverse stability

With $D = 0$ and no governors the models have an exact rotation symmetry and a neutral frequency mode, which produce two structural zero eigenvalues. We evaluate stability on the transverse quotient that removes them and write $\alpha_\perp$ for the largest real part of the transverse spectrum. A portfolio is STABLE if all transverse eigenvalues satisfy $\text{Re } \lambda \leq -2 \times 10^{-3}$ and UNSTABLE if one has $\text{Re } \lambda \geq 2 \times 10^{-3}$; cases in the band between are unresolved and are never counted as stable.

C. Contextual marginals and modal levels

For a policy θ and candidate set V , the set function is $f(S) = \alpha_\perp(S; \theta)$, $S \subseteq V$. A marginal (1) is *material* if $|\Delta_i f(S)| \geq \tau_{\text{mat}} = 0.01 \text{ s}^{-1}$, about 0.2 % damping ratio at 0.7 Hz and 8 % of the flagship margin; its sign class is -1 (stabilizing), 0 or $+1$ (destabilizing). A context S is admissible if S is STABLE.

A change of $\alpha_\perp$ can be a jump of the rightmost eigenvalue from one mode to another, including fast converter-control modes. We therefore grade each marginal (S, i) :

- S is STABLE (global $\alpha_\perp$);
- in addition, the critical mode of S has a match in $S \cup \{i\}$ (bus-voltage shape $\text{MAC} \geq 0.8$, $|\Delta f| \leq 0.15$ Hz) that is the critical mode of $S \cup \{i\}$;
- in addition, both critical frequencies lie in the electromechanical (EM) band 0.1–2.0 Hz;
- (for pairs of contexts) C holds and the two contexts' critical modes match each other.

Under level C, $\Delta_i f(S)$ is the displacement of one tracked EM mode that remains rightmost.

III. WHY NODE WEAKNESS IS NOT INTRINSIC

For $S \subseteq V$ and distinct $i, j \notin S$ let $d_{ij}(S) = f(S \cup \{i, j\}) - f(S \cup \{i\}) - f(S \cup \{j\}) + f(S)$, the one-step second difference. Directly, $d_{ij}(S) = \Delta_i f(S \cup \{j\}) - \Delta_i f(S)$. Recall that f is submodular if $\Delta_i f(S) \geq \Delta_i f(T)$ for $S \subseteq T$, $i \notin T$, equivalently if $d_{ij}(S) \leq 0$ everywhere, and supermodular with the inequalities reversed [29]–[32]. None of the following algebra is new; it is used to turn observed reversals into certificates.

Lemma 1 (chain identity). *Let $S_0 \subseteq V$, $j_1, \dots, j_m \in V \setminus S_0$ distinct, $S_r = S_{r-1} \cup \{j_r\}$, and $i \notin S_m$. Then*

$$\Delta_i f(S_m) = \Delta_i f(S_0) + \sum_{r=1}^m d_{ij_r}(S_{r-1}). \quad (2)$$

Proof: $d_{ij_r}(S_{r-1}) = \Delta_i f(S_r) - \Delta_i f(S_{r-1})$; sum over r . ■

A *nested reversal* of i is a pair $S_1 \subsetneq S_2$, $i \notin S_2$, with $\Delta_i f(S_1)$ and $\Delta_i f(S_2)$ in opposite sign classes. Its direction is stabilizing $\rightarrow$ destabilizing ($s \rightarrow d$) if the negative marginal is at S_1 .

Proposition 1. (a) *If f is submodular, no nested reversal of direction $s \rightarrow d$ exists.* (b) *If f is supermodular, none of direction $d \rightarrow s$ exists.* (c) *If nested reversals of both directions occur, f is neither submodular nor supermodular.*

Proof: Under submodularity every term of (2) is nonpositive, so the marginal cannot increase along a growing chain; (b) is symmetric and (c) is the contrapositive. ■

Proposition 2 (witness). *Let $D = \Delta_i f(S_2) - \Delta_i f(S_1)$ and $m = |S_2 \setminus S_1|$. Along every chain from S_1 to S_2 some term satisfies $d_{ij_r}(S_{r-1}) \geq D/m$ if $D > 0$ ($\leq D/m$ if $D < 0$). A material reversal has $|D| \geq 2\tau_{\text{mat}}$, so the witness has $|d| \geq 2\tau_{\text{mat}}/m$.*

Proof: The m terms of (2) sum to D ; the largest is at least the mean. ■

Nestedness matters. On $V = \{i, a, b\}$ take $f(\emptyset) = f(a) = f(b) = f(ab) = 0$, $f(i) = 2$, $f(ai) = -1$, $f(bi) = 1$, $f(abi) = -2$: all six second differences are ≤ 0 , so f is submodular, yet $\Delta_i f(\{a\}) < 0 < \Delta_i f(\{b\})$. A reversal between incomparable contexts is therefore compatible with submodularity; a nested one is not (Proposition 1).

Proposition 3 (preference drift). *For a chain as in Lemma 1 and $i, j \notin S_m$,*

$$[\Delta_i f - \Delta_j f]_{S_0}^{S_m} = \sum_{r=1}^m (d_{ij_r} - d_{jj_r})(S_{r-1}). \quad (3)$$

If $\Delta_i f(S_0) \leq \Delta_j f(S_0) - \tau$ and $\Delta_j f(S_m) \leq \Delta_i f(S_m) - \tau$, every fixed ranking of V errs by at least τ in one of the two contexts, and some term of (3) is at least $2\tau/m$.

Proof: Apply (2) to i and j and subtract; a fixed ranking places i before j or after it in both contexts. ■

Remark 1. If $f(S) = h(|S|) + \sum_{k \in S} w_k$ then $d_{ik} = d_{jk}$ and the ranking by w is exact in every context, whether h is concave, convex or neither. What defeats a fixed ranking is the heterogeneity of curvature across units, not its sign. The two are measured separately below: sign structure by nested reversals, ranking failure by an oracle ranking.

Because $\alpha_{\perp}$ is a maximum over modes, terms of (2) on intermediate portfolios can reflect mode switches even when the two endpoint marginals are level C. We record whether a certifying term exists whose four portfolios are all stable with EM critical modes (an *EM-clean witness*).

IV. DYNAMIC INTERVENTION SENSITIVITY

A. Frozen and total derivatives

Let $w = (x, z, r)$ collect states, bus voltages and the references r that the dispatch semantics frees. Under *scheduled-power-and-voltage* (SPR) semantics the SG and GFL references are re-solved so that every unit keeps its scheduled active power and terminal voltage; the equilibrium is $R(w; a) = 0$ for a parameter a . For a simple critical eigenvalue λ with right and left eigenvectors u, v of A ,

$$\frac{d\lambda}{da} = \frac{v^H (\partial_a A + \partial_w A[\dot{w}]) u}{v^H u}, \quad \dot{w} = -R_w^{-1} R_a, \quad (4)$$

by the implicit function theorem. The first term alone is the *frozen* derivative D^{fro} ; the full expression is the *total* derivative D^{tot} , and $D^{\text{tot}} - D^{\text{fro}}$ is the re-equilibration term. Equation (4) is the feasible-sensitivity construction of [18], [19] applied to

portfolio-specific operating points; no novelty is claimed for it.

Remark 2. If a enters only a controller of a unit whose scheduled output is fixed (the GFL gain g_i , a PLL scale, an AVR gain), the SPR equilibrium does not depend on a , so $\dot{w} = 0$ and $D^{\text{tot}} = D^{\text{fro}}$. Network and operating-point parameters (branch admittances, voltage setpoints, loads) move the equilibrium.

B. Branch interventions and baselines

A branch intervention scales the complete two-port admittance of branch e by γ_e (tap unchanged). The ranking score of branch e in portfolio S is $-d\alpha_{\perp}/d\gamma_e$ from (4) at $\gamma_e = 1$. The truth is the finite, fully re-solved change $-\alpha_{\perp}(\gamma_e) - \alpha_{\perp}(1)$ at $\gamma_e \in \{1.10, 1.25, 1.50\}$. The static baselines are: branch active and apparent flow $|P_e|, |S_e|$; impedance $|z_e|$; the endpoint electrical distance on the subtransient bus impedance matrix; effective resistance and Fiedler edge score $(u_{2,f} - u_{2,t})^2$ on the coupling Laplacian L_B ; weighted edge betweenness; endpoint dV/dQ ; and the change of the generalized SCR of the target portfolio under $\gamma_e = 1.5$. We also report the conventional sensitivity of the base (all-SG) critical mode, and an independent port-form derivative [11] computed from the network return difference at the critical frequency.

V. EXPERIMENTAL DESIGN

A. Policies and holdouts

Policies lie in the domain $u \in [0, 1]$ ($g = u^2$), $k \in [0.5, 2.3]$, $t \in [0.5, 3.0]$, $h \in [0, 2]$. The *discovery* set has 15 points on axis-aligned lines through the flagship policy $P_4 = (0.03625, 1.425, 1.5, 1)$. An *old holdout* of 24 Latin-hypercube points was used in a first campaign. The *new holdout* HARDENING_H01-H24 is a maximin Latin hypercube (best of 2000 candidate designs by minimum pairwise distance, seed 20260929), frozen before any computation. No point was chosen, replaced or screened for stability. Policies whose all-SG base is unstable (5 new, 6 old) are kept and excluded only from tests that require a stable base.

B. Parameter envelopes

Four envelopes perturb machine and converter parameters within declared bounds: fleet-wide machine groups (EM-f), unit-level machine groups (EM-u), converter groups at every candidate (EC), and both (EMC). A fresh set of ten Latin-hypercube draws per envelope (seed 20260930) is evaluated at P_4 . These are designed conditions, so we report coverage over the declared draws, not probabilities.

C. Computations

At every policy all 512 portfolios of V_9 are solved with bus-voltage mode shapes (32256 portfolio evaluations over 63 policies). Branch sensitivities are computed at 88 new-holdout conditions (24 policies for targets V_4 and V_9 ; 40 draws for V_4) for all 46 branches with three finite steps each. The implicit-function derivatives agree with small-step finite differences within 5% (or 10^{-4} absolute) for 100% of 4048 branch cases. Topology is evaluated at 16 policies for 35 admissible outages and 46 doublings. Reruns of selected tasks were byte-identical.

TABLE II
COVERAGE OF REVERSAL ACROSS THE TESTED BASE-STABLE POLICIES.
LEVEL C IS THE PREREGISTERED PRIMARY GATE (≥ 0.75 ON THE NEW
HOLDOUT).

reversal type	discovery	old holdout	new holdout
any stable-context pair	15/15	18/18	19/19
A: nested, global $\alpha_{\perp}$	15/15	18/18	19/19
B: nested, tracked mode	14/15	16/18	18/19
C: nested, EM same mode	14/15	16/18	17/19
D: C + same family	14/15	16/18	17/19

Both directions ($s \rightarrow d$ and $d \rightarrow s$) at level C occur in 7/19 new-holdout policies. Base-unstable policies (kept, excluded from these denominators): discovery 0, old 6, new 5.

D. Cross-model holdout

No validated grid-forming model exists in our code base. The one existing validated alternative dynamic converter is a WECC library GFL chain (PLL, REGC, REEC-B and REPC-A blocks [33] in ANDES [34]) with a parameter set frozen and validated against an EMT implementation in earlier work. We attach it, unchanged and in constant- Q mode, to a transcription of our machine models that reproduces the internal $\alpha_{\perp}$ of the all-SG base within $2 \times 10^{-7} \text{ s}^{-1}$ at every tested policy (ALT-WECC). It has no g , so policies act through the AVR coordinates. Eight policies were fixed in advance.

E. Statistics

Holdout policies and envelope draws are designed points, so intervals are percentile bootstraps over the tested conditions (policies, or envelopes, as clusters; 10^4 resamples), read as the stability of a summary across the design. Three secondary tests form a Holm family; all gates are effect-size thresholds fixed in the preregistration.

VI. RESULTS

A. Contextual sign reversal

Fig. 1 shows the strongest level-C nested reversal on the new holdout. At policy N24, replacing unit 33 in $S_1 = \{31, 34, 38\}$ moves the tracked 0.1–2 Hz mode left by 0.042 s^{-1} ; in $S_2 = \{31, 34, 37, 38\} \supset S_1$ the same replacement moves the same mode right by 0.064 s^{-1} . Table II gives coverage by level. Every tested policy has a stable-context and a nested reversal of $\alpha_{\perp}$. At the preregistered primary level C, 17/19 base-stable new-holdout policies have one (coverage 0.89; bootstrap interval 0.74–1.00), against 16/18 on the old holdout; the gate (≥ 0.75) passes. Every policy with a level-C reversal also has one at level D, in which the two contexts' critical modes are themselves the same tracked family.

The effects are modest. Level-C reversal magnitudes $\min(|\Delta_i f(S_1)|, |\Delta_i f(S_2)|)$ have quartiles 0.011, 0.013 and 0.017 s^{-1} over the 3263 new-holdout level-C pairs, just above τ_{mat} . Destabilizing $\rightarrow$ stabilizing pairs (2172) outnumber the reverse (1091), and both directions occur within the same policy in 0.37 of the policies.

Fig. 2 maps the fraction of stable contexts in which each replacement destabilizes. No unit is uniformly stabilizing or destabilizing, and the pattern moves with the policy.

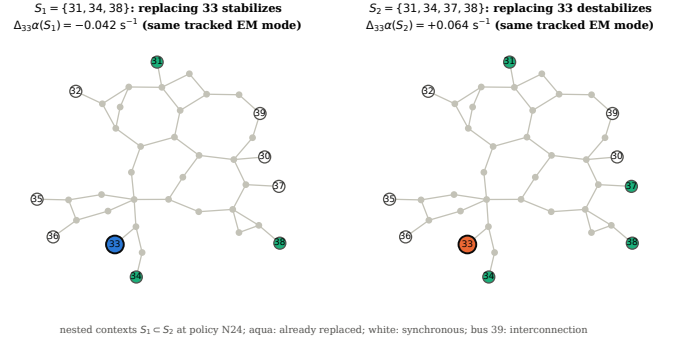


Fig. 1. The same replacement in two nested stable portfolios at a new-holdout policy. Unit 33 moves the tracked electromechanical mode left in S_1 and right in $S_2 \supset S_1$.

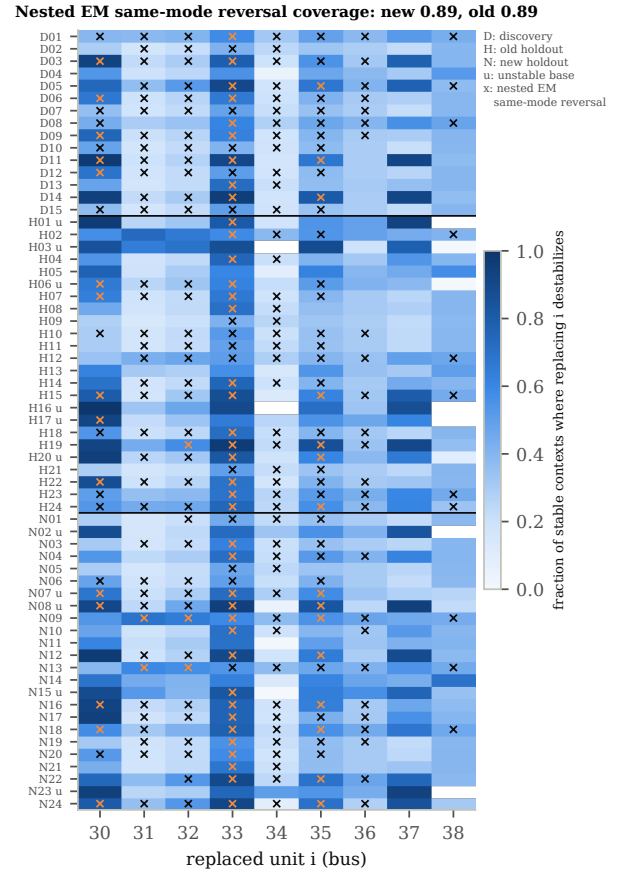


Fig. 2. Fraction of stable contexts in which replacing unit i destabilizes (color), by policy. Crosses: level-C nested reversal. “u”: unstable base.

B. Curvature certificates

Along the canonical chain from S_1 to S_2 , every one of the 3263 new-holdout level-C reversals has a certifying curvature term satisfying the bound of Proposition 2 (Fig. 4); for 94% of them an EM-clean witness exists. Over all one-step squares on the new holdout, second differences exceed $+\tau_{\text{res}}$ (10^{-3} s^{-1}) and fall below $-\tau_{\text{res}}$ in comparable proportions (Fig. 4). By Proposition 1(c), $\alpha_{\perp}$ is neither submodular nor supermodular at every tested policy with reversals of both directions. The

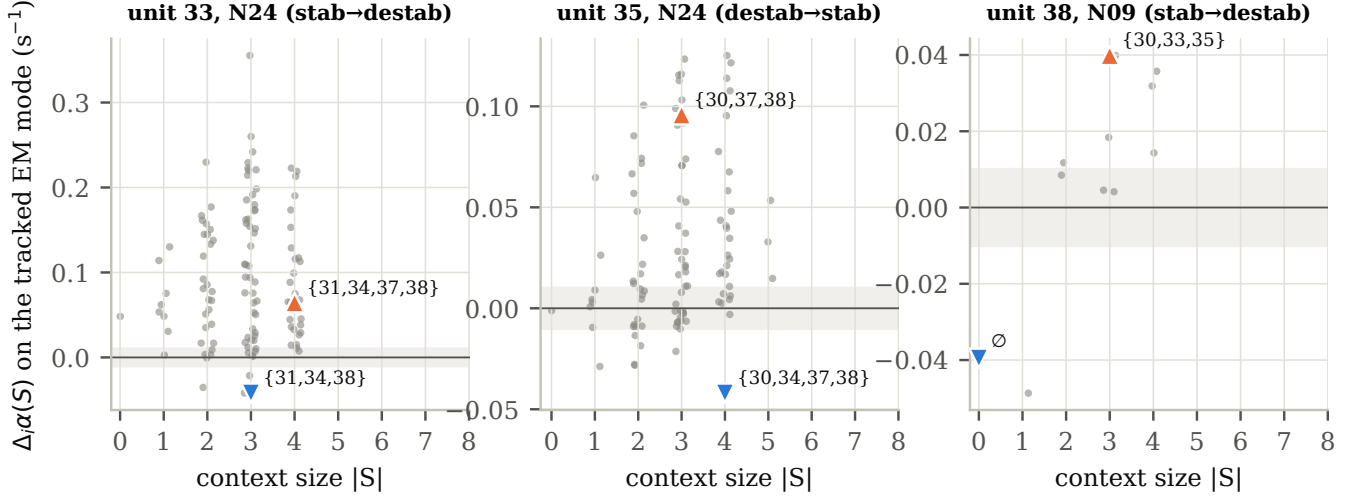


Fig. 3. Strongest level-C nested reversals on base-stable new-holdout policies (three distinct units). Gray: all level-C contexts of the unit; triangles: the nested pair; band: $\pm\tau_{\text{mat}}$.

heavy tails of Fig. 4 ($|d|$ up to 10^4 s^{-1}) are fast-mode events, which is why the certificates are stated on level-C endpoints.

C. Fixed rankings and their modal anatomy

For each base-stable policy, the fixed ranking of the nine units that maximizes agreement with the policy’s own resolved preferences is found exactly by dynamic programming. This is the most favorable case for a node-only ranking. Table III and Fig. 5 give the results. Over all transitions, the oracle ranking orders 0.75 of resolved pairs on the new holdout (interval 0.74–0.78) and its next-replacement choice is at least τ_{mat} worse than the context-aware choice in 0.38 of stable contexts; 19/19 policies have $p^* < 0.90$. The old-to-new shift is -0.002 in p^* (interval -0.036 to 0.041) and 0.007 in regret rate. Rankings do not transfer: a ranking fitted at one policy and used at another is materially wrong in 0.40 of decisions (median over test policies), and 24 distinct optimal orders appear among the 52 base-stable policies.

The stratification changes the reading. Restricted to transitions whose critical modes stay in the EM band, the oracle ranking orders 0.97 of pairs and errs materially in 0.03 of decisions; on level-C marginals the figures are 0.98 and 0.02. In 2089 of the 2265 materially wrong decisions on the new holdout (92%), the unit chosen by the ranking moves the rightmost eigenvalue out of the EM band, almost always onto a fast converter-control mode. The fixed-ranking failure therefore comes mainly from replacements that trigger such a jump in some contexts and not in others. A fixed ranking cannot anticipate these jumps. For EM dynamics alone it is a reasonable approximation, and the reversals of Section VI-A are too small and too rare to change most EM decisions.

Table IV collects the preregistered evidence layers. Seven of eight are positive, including nested reversal (L2) and fixed-ranking failure (L5); the cross-model layer (L8) is not (Section VI-G).

TABLE III
ORACLE-OPTIMAL FIXED NODE RANKING (MEDIAN OVER BASE-STABLE POLICIES). A3 BAR: $p^* \leq 0.90$ AND REGRET RATE ≥ 0.10 .

stratum	old holdout		new holdout	
	p^*	regret	p^*	regret
all transitions	0.75	0.37	0.75	0.38
EM $\rightarrow$ EM transitions	0.97	0.03	0.97	0.03
EM same-mode (level C)	0.98	0.02	0.98	0.02

TABLE IV
PREREGISTERED EVIDENCE LAYERS FOR “A CONTEXT-INDEPENDENT NODE RANKING DOES NOT SUFFICE” (NEW HOLDOUT). STRONG CLAIM: ≥ 5 OF 8 INCLUDING L2 AND L5.

layer	value	pos.
L1 arbitrary-context reversal	1.00	yes
L2 nested-context reversal (level A)	1.00	yes
L3 same-mode nested reversal (level B)	0.95	yes
L4 EM same-mode nested reversal (level C)	0.89	yes
L5 optimal fixed ranking failure (FULL)	$p^*=0.749$; regret=0.376	yes
L6 ranking transfer failure	offdiag regret=0.402; diag=0.376	yes
L7 envelope robustness (fresh draws)	EC: 1.0, EM-f: 0.9, EM-u: 0.8, EMC: 1.0	yes
L8 dynamic-converter holdout (ALT-WECC reversal)	0.0 (MODEL-SPECIFIC)	no

D. Portfolio-conditioned reinforcement ranking

Table V and Fig. 6 report the branch ranking on the new holdout for the core portfolio V_4 (the P4 incompatibility). On the 8 preregistration-eligible conditions (equilibrium residual $\leq 10^{-8}$; the threshold, set before the computation, is below the solver’s typical residual of 10^{-7}), the median Spearman correlation with the finite $\times 1.5$ outcome is 0.985 for D^{tot} , 0.913 for D^{fro} and 0.491 for $|P_e|$, the static index selected on discovery. The median paired advantage over $|P_e|$ is 0.492 (interval 0.477–0.608), and the median top-5 precision of D^{tot} is 1.000. The preregistered gate (lower bound > 0.20 , median $\rho \geq 0.80$, top-5 ≥ 0.70) passes. Over all 64 solved

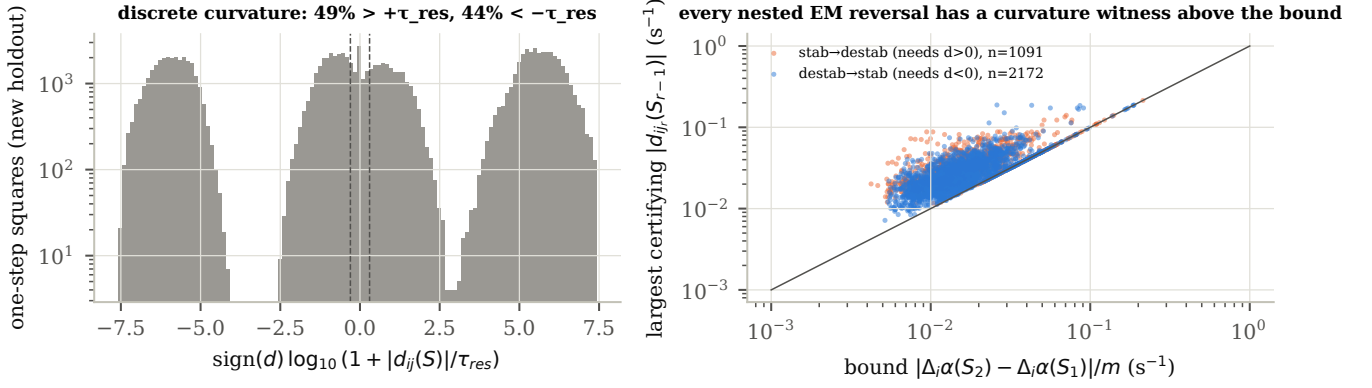


Fig. 4. Left: signed, log-compressed one-step second differences over the new holdout. Right: for each level-C nested reversal, the largest certifying curvature term against the bound of Proposition 2.

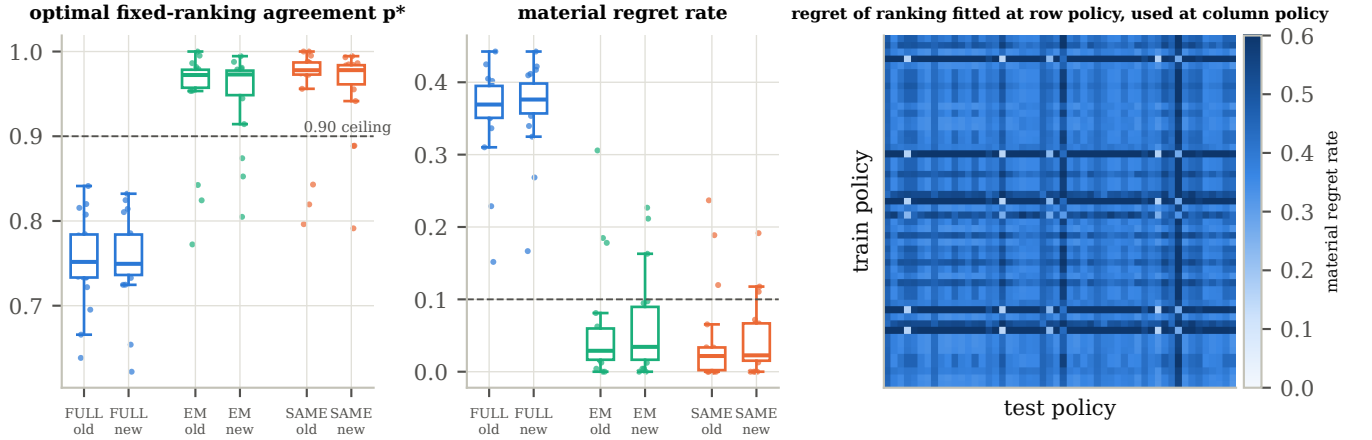


Fig. 5. Oracle fixed-ranking agreement p^* and material regret rate by stratum (old and new holdout), and the regret of rankings transferred between policies (rows: policy of fit; columns: policy of use).

TABLE V
BRANCH-REINFORCEMENT RANKING ON THE NEW HOLDOUT (TARGET H4, TRUTH: FINITE RE-EQUILIBRATED $\times 1.5$). MEDIAN OVER CONDITIONS; “ELIG.”: THE 8 PREREGISTRATION-ELIGIBLE CONDITIONS ($R_0 \leq 10^{-8}$); OTHER COLUMNS: ALL 64 SOLVED CONDITIONS.

predictor	ρ elig.	ρ	τ	concord.	top-5	NDCG@5
$ P_e $ (S1, selected)	0.49	0.50	0.39	0.71	0.20	0.61
$ S_e $	0.47	0.48	0.37	0.69	0.20	0.61
$ z_e $	0.25	0.24	0.20	0.61	0.20	0.27
electrical distance	0.21	0.19	0.15	0.58	0.40	0.38
effective resistance	0.30	0.30	0.22	0.61	0.60	0.58
Fiedler edge score	-0.05	-0.05	-0.03	0.48	0.20	0.36
edge betweenness	-0.28	-0.27	-0.18	0.40	0.00	0.17
endpoint dV/dQ	-0.17	-0.16	-0.10	0.45	0.00	0.16
$\Delta gSCR(H4)$	0.16	0.17	0.12	0.57	0.20	0.30
base-portfolio eig. sens.	-0.05	0.03	0.01	0.50	0.60	0.91
frozen (portfolio)	0.91	0.90	0.77	0.90	0.60	0.97
total (portfolio)	0.98	0.99	0.93	0.98	1.00	0.99
port form (policies only)	0.99	1.00	0.96	1.00	1.00	1.00

conditions (residual $\leq 2.1 \times 10^{-7}$) the figures are 0.987, 0.895 and 0.497, with paired advantage 0.489 (interval 0.484–0.638). D^{tot} outranks every static index in every condition, including the best one chosen after the fact per condition.

The advantage holds in each stratum: EM-band conditions

(0.988 vs 0.498), tracked-mode truth (0.988), smaller steps ($\gamma = 1.10$: 0.999; 1.25: 0.995), and the nine-unit target V_9 , where $|P_e|$ is negatively correlated with the outcome (-0.183 vs 0.994). The sensitivity of the *base* portfolio has median ρ 0.034: the branch ranking belongs to the portfolio, not to the network. The port-form derivative agrees with D^{tot} . Fig. 6 (right) shows that the first-order prediction overstates the largest reinforcements, whose effect saturates at $\gamma = 1.5$; the ranking is unaffected.

E. When re-equilibration matters

Splitting (4) into frozen and re-equilibration parts (Fig. 7), the controller coordinates behave as the remark predicts: for g_i , PLL scales and AVR gains the two derivatives agree to $2.2e-07$ relative. For loads the re-equilibration term exceeds the frozen term in 61% of cases, and for voltage setpoints the frozen term is identically zero. For branches the re-equilibration term exceeds the frozen one for about half of the branches; it changes the ranking materially (sign disagreement $\geq 10\%$ or Kendall ≤ 0.6) in 28% of the V_4 conditions, and every condition has at least one branch whose frozen and total signs differ. The effect is largest across policies: on the 24 new

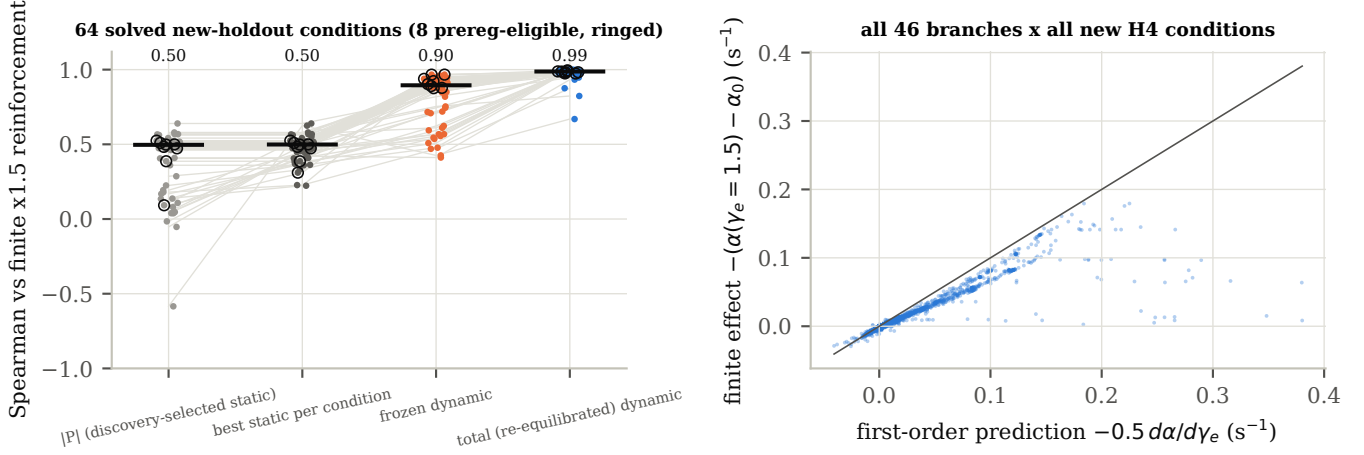


Fig. 6. Left: Spearman correlation with the finite reinforcement outcome per new-holdout condition (lines join the same condition; ringed: the preregistration-eligible subset). Right: first-order prediction vs finite effect, all branches and conditions.

policies alone the frozen derivative's median correlation drops to 0.712, while D^{tot} stays at 0.996.

F. Robustness

Under the fresh envelope draws at P4, the coverage of a stable-context reversal on the core lattice is 1.00 (EC), 0.90 (EM-f), 0.80 (EM-u) and 1.00 (EMC); for EM same-mode reversals it is 0.00, 0.50, 0.60 and 0.70. Converter-parameter draws alone (EC) remove the EM reversals from the four-unit lattice while preserving the global ones. For branch ranking the fresh draws give median ρ 0.986 for D^{tot} against 0.503 for $|P_e|$.

G. Cross-model holdout

Table VI summarizes ALT-WECC. The model passes its qualification (base $\alpha_{\perp}$ reproduced, 488 of 488 cases initialized with residual $\leq 3 \times 10^{-13}$, no active limit). Its converted portfolios are much less stable than with the custom GFL: V_4 is unstable at all eight policies, with 0.3–0.8 Hz critical modes. No stable-context reversal occurs: in every stable core context, every replacement leaves $\alpha_{\perp}$ unchanged or increases it. The frozen chain regulates constant reactive power and cannot contribute voltage support, which is consistent with, but does not establish, the reading that the stabilizing branch of the reversal requires converter voltage control. The finite branch ranking is model-specific (median Kendall 0.02 between the two models on 12 preregistered branches), as a previous cross-tool check had indicated. The method transfers: within ALT-WECC, a re-solved finite-difference total sensitivity ranks the model's own finite reinforcements with median ρ 0.99 against 0.79 for $|P_e|$. The top equal-budget corridor and the direction of the six preregistered topology actions agree in 0.75 of policies and 0.94 of material pairs.

H. Topology, electromechanical modes only

Topology effects were re-evaluated by tracking the pre-action EM-band rightmost mode of each portfolio through

TABLE VI
CROSS-MODEL HOLDOUT: FROZEN TX3 WECC LIBRARY GFL CHAIN (ALT-WECC, ANDES) VS THE CUSTOM GFL, 8 PREREGISTERED POLICIES.

test	value	verdict
A stable-context reversal, core lattice	0.00 of 7 (custom 0.71)	model-specific
B same-mode reversal	0.00	model-specific
C 12-line finite ranking, Kendall custom vs ALT	median 0.02	model-specific
C2 ALT total vs ALT finite, minus $ P $	0.99 vs 0.79	transfers
D corridor top-1 agreement	0.75	transfers
E topology sign agreement	0.94 (35 pairs)	transfers

every admissible action (Fig. 8). Events in which the full $\alpha_{\perp}$ is set by a mode outside 0.1–2 Hz and differs from the tracked EM change are labelled fast-mode dominated (50 of the V_4 cases) and excluded. At P4, doubling any one of lines 1–2, 1–39, 8–9 or the 23–36 transformer makes V_4 stable, removing an EM incompatibility; across the 16 policies, 45 (policy, action) pairs create an EM-failing hyperedge of at most three units where none existed, and no creation is fast-mode only. Against the tracked EM effect, no static topology score meets the prediction rule ($|\rho| \geq 0.6$ in $\geq 75\%$ of policies): the changes of the Fiedler value, Kirchhoff index, gSCR of V_4 and minimum core SCR have median ρ -0.50, 0.56, -0.54 and -0.54, reaching $|\rho| \geq 0.6$ in at most 0.38 of policies. These scores carry partial information—mostly the distinction between outages and doublings—but do not order actions within that distinction.

I. A negative result on modal mixing

A first campaign reported that controller-weighted modal mixing of the network port operator correlates with the number of reversing units ($\rho = 0.66$ on the old holdout). That quantity was evaluated at each policy's own critical frequency, which confounds controller effects with frequency movement. At a fixed reference frequency the old-holdout correlation falls to

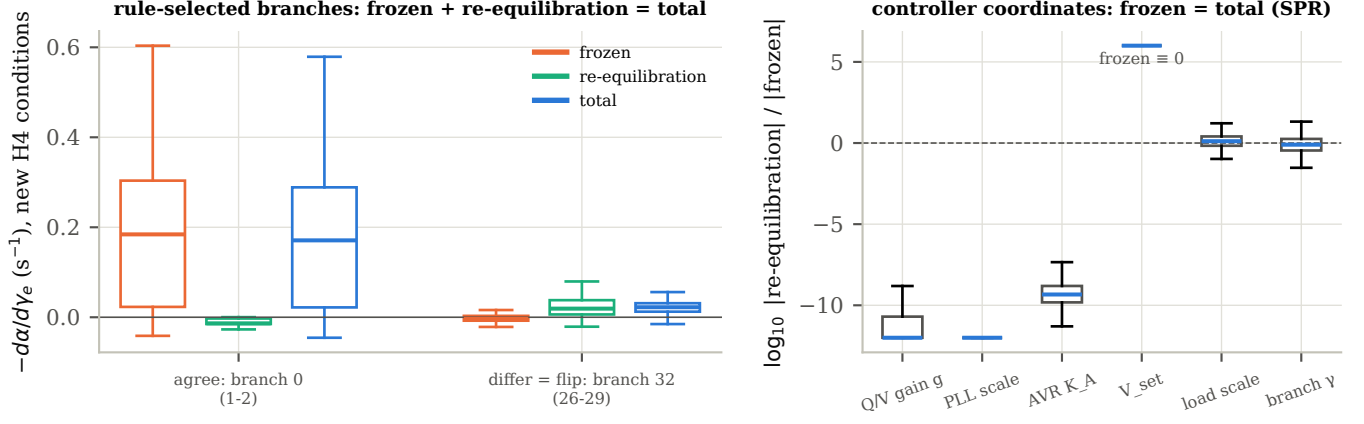


Fig. 7. Left: frozen, re-equilibration and total branch sensitivities for the branches selected by the preregistered rule (largest-effect branch with agreeing ranks; branch with the largest rank displacement, which is also the most frequent sign flip). Right: ratio of the re-equilibration term to the frozen term by coordinate class.

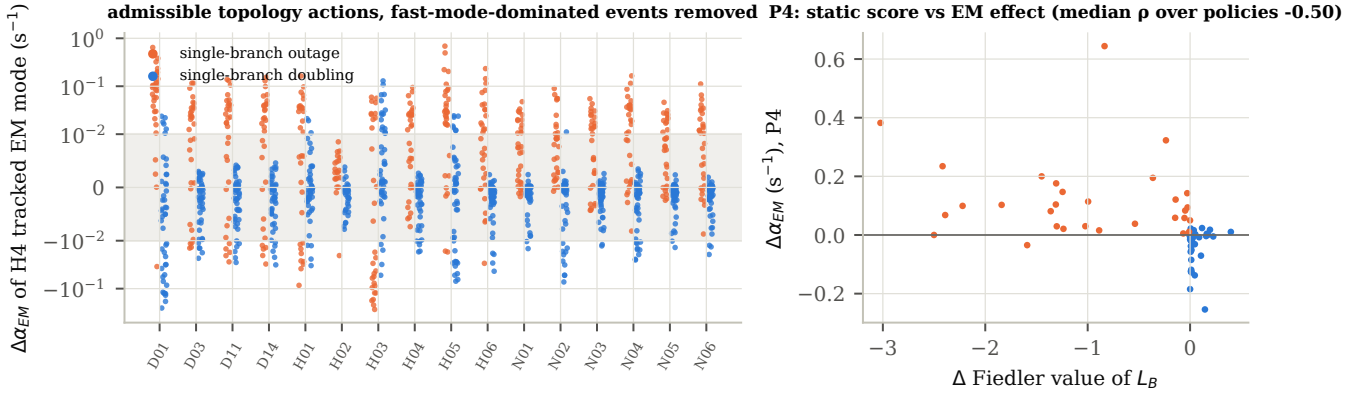


Fig. 8. Left: tracked EM-mode change of V_4 for every admissible action, by policy (fast-mode-dominated events removed; band: $\pm\tau_{mat}$). Right: at P4, EM effect against the change of the Fiedler value.

0.21. On the new holdout the preregistered fixed-frequency test gives $\rho = 0.55$ (permutation $p = 0.016$, Holm-adjusted 0.033), which misses the preregistered bar ($p \leq 0.01$). An exact two-factor decomposition attributes 19% of the variance of the mixing change to the controller term and the rest to frequency movement. The mixing interpretation of contextuality is not supported.

VII. DISCUSSION

Relation to strength metrics. Grid-strength indices are properties of the network and of a location. The results indicate that two different objects are being asked of them: a location's support for converters, which such indices approximate, and the ranking of the next intervention in a given portfolio, which depends on the portfolio. The static indices we tested are not uninformative—effective resistance puts some of the strongest branches in its top five, and the distinction between outages and doublings is visible in the Fiedler value—but none orders interventions reliably within a portfolio. Placement methods built on grid-strength or Kron-reduced-Laplacian surrogates treat converters as homogeneous and, for grid-forming conversions, can prove that each conversion

helps [35]–[37]; the reversals reported here arise in exactly the setting those surrogates abstract away, heterogeneous machines and converters with their own controls. Offline priority lists of switching actions [38] are fixed rankings of the kind examined in Section VI-C.

Relation to eigenvalue sensitivity. Equation (4) is classical, and a recent preprint [22] chains impedance-based sensitivities through the power-flow Jacobian to line admittances. What we add is the evaluation of the portfolio-conditioned total derivative as a *ranking*, on held-out policies and parameter draws, against finite re-solved reinforcements at three magnitudes and against static indices chosen in advance and after the fact; and the decomposition that locates the re-equilibration term in network and operating-point coordinates.

Planning interpretation. Two practical consequences follow. First, a replacement sequence evaluated with a fixed weak-bus list will choose a materially worse next unit in about a third of stable contexts on this benchmark, mostly because the list cannot anticipate which replacements push the system onto fast converter-control modes. Second, the reinforcement ranking should be recomputed for the portfolio in which the reinforcement is applied; its cost is one factorization of the

equilibrium Jacobian per portfolio and one back-substitution per candidate branch, instead of a re-solved equilibrium and eigenanalysis per candidate.

What the Laplacian does not capture. Earlier analyses on the same benchmark found that the graph modes carrying the critical mode's energy are not the lowest Laplacian modes (Jaccard overlap 0.2 with the three lowest) and that witness changes rarely coincide with modal-support changes (3 of 9 events). With the mixing result above, the evidence points to the same conclusion: the contextual structure lives in the controlled operator, not in the static spectrum of the network.

Literature scope. The gap statements rest on a focused search of 2020–2026 journal and conference records (57 verified entries, arXiv only where no peer-reviewed version exists) without full-text database access; they are stated as the absence of such results within that search.

Limitations. All results are phasor-domain small-signal results on one benchmark with one custom GFL model; the cross-model holdout shows that reversals and branch rankings are converter-model-specific, and no grid-forming model was tested. Converter limits are not modeled, and the fast control-mode instabilities that drive the fixed-ranking failure are those of an idealized converter. Reversal magnitudes are a few hundredths of s^{-1} . Policies and envelopes are designed points; no probability statement is made. The preregistered equilibrium-residual eligibility threshold was stricter than the solver tolerance, which reduced the primary GOLD-B set to 8 conditions; the full set gives the same conclusion. No electromagnetic-transient validation was performed.

VIII. CONCLUSION

On the IEEE 39-bus benchmark, the marginal small-signal effect of replacing a synchronous generator by a grid-following converter depends on the portfolio in which the replacement occurs. Nested reversals on the same electromechanical mode recur on a preregistered holdout and certify discrete curvature of both signs. An oracle fixed ranking of units selects a materially worse next replacement in about a third of stable contexts, a failure driven mainly by fast converter-control modes. The portfolio-conditioned re-equilibrated branch sensitivity ranks finite reinforcements with median Spearman near 0.99 where static indices reach about 0.5. With a frozen library GFL chain the ranking method transfers, while the reversals and the specific ranking do not. Weakness indices should be read as properties of a network; intervention rankings should be computed for the portfolio in which they are used.

DATA AND CODE AVAILABILITY

Preregistration, deviation log, code, result tables and figure sources are in the project repository, branch `research/contextual-dynamic-weakness-hardening`.

REFERENCES

- [1] D. Wu, G. Li, M. Javadi, A. M. Malyscheff, M. Hong, and J. N. Jiang, "Assessing Impact of Renewable Energy Integration on System Strength Using Site-Dependent Short Circuit Ratio," *IEEE Transactions on Sustainable Energy*, vol. 9, no. 3, pp. 1072–1080, 2018.
- [2] H. Xin, W. Dong, X. Yuan, D. Gan, K. Wang, and H. Xie, "Generalized short circuit ratio for multi power electronic based devices infeed to power systems," *Proceedings of the CSEE*, vol. 36, no. 22, pp. 6013–6027, 2016.
- [3] W. Dong, H. Xin, D. Wu, and L. Huang, "Small Signal Stability Analysis of Multi-Infeed Power Electronic Systems Based on Grid Strength Assessment," *IEEE Transactions on Power Systems*, vol. 34, no. 2, pp. 1393–1403, 2019.
- [4] Y. Zhou, H. Xin, D. Wu, F. Liu, Z. Li, G. Wang, H. Yuan, and P. Ju, "Small-Signal Stability Assessment of Heterogeneous Grid-Following Converter Power Systems Based on Grid Strength Analysis," *IEEE Transactions on Power Systems*, vol. 38, no. 3, pp. 2566–2579, 2023.
- [5] C. Liu, H. Xin, D. Wu, H. Gao, H. Yuan, and Y. Zhou, "Generalized Operational Short-Circuit Ratio for Grid Strength Assessment in Power Systems With High Renewable Penetration," *IEEE Transactions on Power Systems*, vol. 39, no. 4, pp. 5479–5494, 2024.
- [6] F. Ma, H. Xin, D. Wu, Y. Liu, and X. Chen, "Assessing Small-Signal Grid Strength of 100% Inverter-Based Power Systems," *IEEE Transactions on Power Delivery*, vol. 39, no. 5, pp. 2784–2796, 2024.
- [7] C. Henderson, A. Egea-Alvarez, T. Kneuppel, G. Yang, and L. Xu, "Grid Strength Impedance Metric: An Alternative to SCR for Evaluating System Strength in Converter Dominated Systems," *IEEE Transactions on Power Delivery*, vol. 39, no. 1, pp. 386–396, 2024.
- [8] Y. Lamrani, T. Prevost, C. Cardozo, C. Zanabria, F. Colas, and X. Guillaud, "Frequency-Averaged Grid Impedance Metric for Identifying Weakest Buses in Power Electronics-Dominated Systems," in *2025 IEEE Kiel PowerTech*, 2025, pp. 1–7.
- [9] E. Ebrahimzadeh, F. Blaabjerg, X. Wang, and C. L. Bak, "Bus Participation Factor Analysis for Harmonic Instability in Power Electronics Based Power Systems," *IEEE Transactions on Power Electronics*, vol. 33, no. 12, pp. 10 341–10 351, 2018.
- [10] Y. Zhu, Y. Gu, Y. Li, and T. Green, "Participation Analysis in Impedance Models: The Grey-Box Approach for Power System Stability," *IEEE Transactions on Power Systems*, vol. 37, no. 1, pp. 343–353, 2022.
- [11] Y. Zhu, Y. Gu, Y. Li, and T. C. Green, "Impedance-Based Root-Cause Analysis: Comparative Study of Impedance Models and Calculation of Eigenvalue Sensitivity," *IEEE Transactions on Power Systems*, vol. 38, no. 2, pp. 1642–1654, 2023.
- [12] T. Coletta and P. Jacquod, "Linear stability and the Braess paradox in coupled-oscillator networks and electric power grids," *Physical Review E*, vol. 93, no. 3, 2016.
- [13] B. Schäfer, T. Pesch, D. Manik, J. Gollenstede, G. Lin, H.-P. Beck, D. Witthaut, and M. Timme, "Understanding Braess' Paradox in power grids," *Nature Communications*, vol. 13, no. 1, 2022.
- [14] Y. Song, D. J. Hill, and T. Liu, "Network-Based Analysis of Small-Disturbance Angle Stability of Power Systems," *IEEE Transactions on Control of Network Systems*, vol. 5, no. 3, pp. 901–912, 2018.
- [15] Y. Li, T. C. Green, and Y. Gu, "The Intrinsic Communication in Power Systems: A New Perspective to Understand Synchronization Stability," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 70, no. 11, pp. 4615–4626, 2023.
- [16] U. Markovic, O. Stanojev, P. Aristidou, E. Vrettos, D. Callaway, and G. Hug, "Understanding Small-Signal Stability of Low-Inertia Systems," *IEEE Transactions on Power Systems*, vol. 36, no. 5, pp. 3997–4017, 2021.
- [17] O. Stanojev, U. Markovic, P. Aristidou, and G. Hug, "Improving Stability of Low-Inertia Systems Using Virtual Induction Machine Synchronization for Grid-Following Converters," *IEEE Transactions on Power Systems*, vol. 38, no. 3, pp. 2290–2303, 2023.
- [18] T. Smed, "Feasible eigenvalue sensitivity for large power systems," *IEEE Transactions on Power Systems*, vol. 8, no. 2, pp. 555–563, 1993.
- [19] H.-K. Nam, Y.-K. Kim, K.-S. Shim, and K. Lee, "A new eigen-sensitivity theory of augmented matrix and its applications to power system stability analysis," *IEEE Transactions on Power Systems*, vol. 15, no. 1, pp. 363–369, 2000.
- [20] S. Mendoza-Armenta and I. Dobson, "Applying a Formula for Generator Redispatch to Damp Interarea Oscillations Using Synchrophasors," *IEEE Transactions on Power Systems*, vol. 31, no. 4, pp. 3119–3128, 2016.
- [21] Y. Li, W. Li, X. Shi, and X. Chen, "An efficient numerical method for computing eigenvalue sensitivity with respect to operational parameters of large-scale power systems," *Electric Power Systems Research*, vol. 174, p. 105859, 2019.
- [22] T. Joswig-Jones, W. Dong, B. Tan *et al.*, "Sensitivity-based system strength assessment: Mapping power flow and network topology perturbations to system eigenvalues," arXiv:2607.28764 (preprint, not peer reviewed), 2026.

- [23] A. T. Saric and A. M. Stankovic, "Rapid Small-Signal Stability Assessment and Enhancement Following Changes in Topology," *IEEE Transactions on Power Systems*, vol. 30, no. 3, pp. 1155–1163, 2015.
- [24] C. Li, H.-D. Chiang, and Z. Du, "Online Line Switching Method for Enhancing the Small-Signal Stability Margin of Power Systems," *IEEE Transactions on Smart Grid*, vol. 9, no. 5, pp. 4426–4435, 2018.
- [25] T. H. Summers, F. L. Cortesi, and J. Lygeros, "On Submodularity and Controllability in Complex Dynamical Networks," *IEEE Transactions on Control of Network Systems*, vol. 3, no. 1, pp. 91–101, 2016.
- [26] A. Olshevsky, "On (Non)Supermodularity of Average Control Energy," *IEEE Transactions on Control of Network Systems*, vol. 5, no. 3, pp. 1177–1181, 2018.
- [27] T. Athay, R. Podmore, and S. Virmani, "A Practical Method for the Direct Analysis of Transient Stability," *IEEE Transactions on Power Apparatus and Systems*, vol. PAS-98, no. 2, pp. 573–584, 1979.
- [28] P. W. Sauer and M. A. Pai, *Power System Dynamics and Stability*. Upper Saddle River, NJ: Prentice Hall, 1998.
- [29] D. M. Topkis, "Minimizing a Submodular Function on a Lattice," *Operations Research*, vol. 26, no. 2, pp. 305–321, 1978.
- [30] L. Lovász, "Submodular functions and convexity," in *Mathematical Programming The State of the Art*. Springer Berlin Heidelberg, 1983, pp. 235–257.
- [31] S. Fujishige, *Submodular Functions and Optimization*, 2nd ed., ser. Annals of Discrete Mathematics. Amsterdam: Elsevier, 2005, vol. 58.
- [32] F. Bach, "Learning with Submodular Functions: A Convex Optimization Perspective," *Foundations and Trends in Machine Learning*, vol. 6, no. 2-3, pp. 145–373, 2013.
- [33] P. Pourbeik, J. J. Sanchez-Gasca, J. Senthil, J. D. Weber, P. Zakhast, Y. Kazachkov, S. Tacke, J. Wen, and A. Ellis, "Generic Dynamic Models for Modeling Wind Power Plants and Other Renewable Technologies in Large-Scale Power System Studies," *IEEE Transactions on Energy Conversion*, vol. 32, no. 3, pp. 1108–1116, 2017.
- [34] H. Cui, F. Li, and K. Tomsovic, "Hybrid Symbolic-Numeric Framework for Power System Modeling and Analysis," *IEEE Transactions on Power Systems*, vol. 36, no. 2, pp. 1373–1384, 2021.
- [35] C. Yang, L. Huang, H. Xin, and P. Ju, "Placing Grid-Forming Converters to Enhance Small Signal Stability of PLL-Integrated Power Systems," *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3563–3573, 2021.
- [36] H. Xin, C. Liu, X. Chen, Y. Wang, E. Prieto-Araujo, and L. Huang, "How Many Grid-Forming Converters Do We Need? A Perspective From Small Signal Stability and Power Grid Strength," *IEEE Transactions on Power Systems*, vol. 40, no. 1, pp. 623–635, 2025.
- [37] L. Huang, H. Xin, W. Dong, and F. Dörfler, "Impacts of Grid Structure on PLL-Synchronization Stability of Converter-Integrated Power Systems," *IFAC-PapersOnLine*, vol. 55, no. 13, pp. 264–269, 2022.
- [38] M. Khaji and M. Aghamohammadi, "Emergency transmission line switching to suppress power system inter-area oscillation," *International Journal of Electrical Power & Energy Systems*, vol. 87, pp. 52–64, 2017.